

Equations

1 Households

The representative household solves:

$$\max_{\{C_t, I_t, K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln C_t \quad (1)$$

subject to the budget constraint:

$$C_t + I_t + T_t = w_t L_t + r_t K_t + \Pi_t, \quad (2)$$

and the law of motion for capital:

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (3)$$

Labor supply is inelastic and fixed:

$$L_t = \bar{L}. \quad (4)$$

The Euler equation is:

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} (1 + r_{t+1} - \delta). \quad (5)$$

In steady state, the Euler equation implies:

$$r = \frac{1}{\beta} - 1 + \delta. \quad (6)$$

Steady-state investment satisfies:

$$I = \delta K. \quad (7)$$

2 Final-Good Producer

The competitive final-good producer combines the outputs of the two intermediate goods sectors into aggregate final output:

$$Y_t = Y_{1,t}^\theta Y_{2,t}^{1-\theta}, \quad 0 < \theta < 1. \quad (8)$$

The final good is the numeraire. The final-good producer solves:

$$\max_{Y_{1,t}, Y_{2,t}} (Y_{1,t}^\theta Y_{2,t}^{1-\theta} - P_{1,t} Y_{1,t} - P_{2,t} Y_{2,t}). \quad (9)$$

The first-order conditions imply the sectoral prices:

$$P_{1,t} = \theta \frac{Y_t}{Y_{1,t}}, \quad (10)$$

$$P_{2,t} = (1 - \theta) \frac{Y_t}{Y_{2,t}}. \quad (11)$$

3 Intermediate Goods Sectors

There are two intermediate goods sectors, indexed by $j \in \{1, 2\}$. Sector j produces:

$$Y_{j,t} = A_j K_{j,t}^{\alpha_j} L_{j,t}^{\beta_j} (\chi_t E_{j,t})^{1-\alpha_j-\beta_j}. \quad (12)$$

Define the energy share parameter:

$$\gamma_j = 1 - \alpha_j - \beta_j. \quad (13)$$

Equivalently:

$$Y_{j,t} = A_j K_{j,t}^{\alpha_j} L_{j,t}^{\beta_j} (\chi_t E_{j,t})^{\gamma_j}. \quad (14)$$

Each intermediate goods firm takes $P_{j,t}$, r_t , w_t , and $P_{E,j,t}$ as given and solves:

$$\max_{K_{j,t}, L_{j,t}, E_{j,t}} (P_{j,t} Y_{j,t} - r_t K_{j,t} - w_t L_{j,t} - P_{E,j,t} E_{j,t}). \quad (15)$$

The first-order conditions are:

$$r_t = P_{j,t} \alpha_j \frac{Y_{j,t}}{K_{j,t}}, \quad (16)$$

$$w_t = P_{j,t} \beta_j \frac{Y_{j,t}}{L_{j,t}}, \quad (17)$$

$$P_{E,j,t} = P_{j,t} \gamma_j \frac{Y_{j,t}}{E_{j,t}}. \quad (18)$$

Under perfect competition, profits in each intermediate goods sector are zero:

$$\Pi_{j,t} = 0. \quad (19)$$

Total intermediate-goods-sector profits are:

$$\Pi_t^Y = \sum_{j=1}^2 \Pi_{j,t}. \quad (20)$$

4 Energy Services

Each intermediate goods sector combines fossil and renewable energy into energy services:

$$E_{j,t} = \left[\omega_j E_{f,j,t}^{\frac{\sigma-1}{\sigma}} + (1 - \omega_j) E_{r,j,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \quad (21)$$

The sector-specific energy price index is:

$$P_{E,j,t} = \left[\omega_j^\sigma P_{f,t}^{1-\sigma} + (1 - \omega_j)^\sigma P_{r,t}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (22)$$

Conditional demand for fossil energy is:

$$E_{f,j,t} = \omega_j^\sigma \left(\frac{P_{f,t}}{P_{E,j,t}} \right)^{-\sigma} E_{j,t}. \quad (23)$$

Conditional demand for renewable energy is:

$$E_{r,j,t} = (1 - \omega_j)^\sigma \left(\frac{P_{r,t}}{P_{E,j,t}} \right)^{-\sigma} E_{j,t}. \quad (24)$$

5 Energy Sector

The energy sector produces fossil and renewable energy by transforming the aggregate final good into energy inputs.

Fossil energy production is:

$$E_{f,t} = \frac{z_f}{c_f} M_{f,t}. \quad (25)$$

Renewable energy production is:

$$E_{r,t} = \frac{z_r}{c_r} M_{r,t}. \quad (26)$$

Equivalently, the final-good inputs required for energy production are:

$$M_{f,t} = \frac{c_f}{z_f} E_{f,t}, \quad (27)$$

$$M_{r,t} = \frac{c_r}{z_r} E_{r,t}. \quad (28)$$

Under perfect competition, fossil and renewable energy prices are equal to marginal costs net of the renewable energy subsidy:

$$P_{f,t} = \frac{c_f}{z_f}, \quad (29)$$

$$P_{r,t} = \frac{c_r}{z_r} - s_t. \quad (30)$$

The subsidy is assumed to satisfy:

$$0 \leq s_t < \frac{c_r}{z_r}. \quad (31)$$

Energy-sector profits are:

$$\Pi_t^E = P_{f,t} E_{f,t} + P_{r,t} E_{r,t} - M_{f,t} - M_{r,t} + s_t E_{r,t}. \quad (32)$$

Using the pricing equations and perfect competition:

$$\Pi_t^E = 0. \quad (33)$$

Total profits rebated to households are:

$$\Pi_t = \Pi_t^Y + \Pi_t^E. \quad (34)$$

6 Government

The government subsidizes renewable energy production. The subsidy is financed by a lump-sum tax on households:

$$T_t = s_t E_{r,t}. \quad (35)$$

7 Resource Constraint

Aggregate final output is used for consumption, investment, and as an input in energy production:

$$Y_t = C_t + I_t + M_{f,t} + M_{r,t}. \quad (36)$$

Using the linear energy production technologies, this can be written as:

$$Y_t = C_t + I_t + \frac{c_f}{z_f} E_{f,t} + \frac{c_r}{z_r} E_{r,t}. \quad (37)$$

8 Market Clearing

Capital market clearing is:

$$K_t = K_{1,t} + K_{2,t}. \quad (38)$$

Labor market clearing is:

$$L_t = L_{1,t} + L_{2,t}. \quad (39)$$

Total labor supply is fixed:

$$L_t = \bar{L}. \quad (40)$$

Fossil energy market clearing is:

$$E_{f,t} = E_{f,1,t} + E_{f,2,t}. \quad (41)$$

Renewable energy market clearing is:

$$E_{r,t} = E_{r,1,t} + E_{r,2,t}. \quad (42)$$

9 Competitive Equilibrium

A competitive equilibrium is a sequence of quantities:

$$\{C_t, I_t, K_t, L_t, K_{1,t}, K_{2,t}, L_{1,t}, L_{2,t}, Y_t, Y_{1,t}, Y_{2,t}, E_{j,t}, E_{f,j,t}, E_{r,j,t}, E_{f,t}, E_{r,t}, M_{f,t}, M_{r,t}\}_{t=0}^{\infty}, \quad j \in \{1, 2\}, \quad (43)$$

and prices:

$$\{w_t, r_t, P_{1,t}, P_{2,t}, P_{f,t}, P_{r,t}, P_{E,j,t}\}_{t=0}^{\infty}, \quad j \in \{1, 2\}, \quad (44)$$

such that:

1. the representative household satisfies its budget constraint, capital accumulation equation, and Euler equation;
2. the final-good producer combines the two sectoral goods into aggregate output and satisfies its first-order conditions;
3. intermediate goods firms maximize profits and satisfy the first-order conditions for capital, labor, and energy demand;
4. intermediate goods firms minimize the cost of obtaining energy services;
5. the energy sector satisfies its production and pricing equations;
6. the government budget constraint holds;
7. goods, capital, labor, fossil energy, and renewable energy markets clear.

10 Steady State

In steady state, all endogenous variables are constant over time. Thus:

$$C_t = C, \quad K_t = K, \quad I_t = I, \quad Y_t = Y, \quad (45)$$

$$Y_{j,t} = Y_j, \quad K_{j,t} = K_j, \quad L_{j,t} = L_j, \quad E_{j,t} = E_j, \quad (46)$$

$$E_{f,j,t} = E_{f,j}, \quad E_{r,j,t} = E_{r,j}, \quad E_{f,t} = E_f, \quad E_{r,t} = E_r. \quad (47)$$

Prices are also constant:

$$w_t = w, \quad r_t = r, \quad P_{1,t} = P_1, \quad P_{2,t} = P_2, \quad (48)$$

$$P_{f,t} = P_f, \quad P_{r,t} = P_r, \quad P_{E,j,t} = P_{E,j}. \quad (49)$$

Policy and technology parameters are constant:

$$s_t = s, \quad \chi_t = \chi. \quad (50)$$

The steady-state Euler equation implies:

$$r = \frac{1}{\beta} - 1 + \delta. \quad (51)$$

Capital accumulation implies:

$$I = \delta K. \quad (52)$$

Aggregate final output is produced according to:

$$Y = Y_1^\theta Y_2^{1-\theta}. \quad (53)$$

The steady-state resource constraint is:

$$Y = C + \delta K + \frac{c_f}{z_f} E_f + \frac{c_r}{z_r} E_r. \quad (54)$$

11 Additional Variables

The renewable energy share is defined as:

$$\text{Renewable Share} = \frac{E_r}{E_f + E_r}. \quad (55)$$

If the output gap is measured relative to a baseline level of output Y^{base} , then:

$$\text{Output Gap} = \frac{Y - Y^{base}}{Y^{base}}. \quad (56)$$